

1. Administrative & Study Identification

Posting ID: 270112071640985543 **Study Title:** A Real-World Study to Evaluate Luspatercept in Adults With Transfusion-Dependent Beta-Thalassemia in the Middle East **Official Title:** REal-World Application of Luspatercept in Adults With Transfusion-Dependent Beta-Thalassemia in the Middle East (RELATE): A Non-interventional Retrospective and Prospective Observational Study **Short Title/Acronym:** RELATE Study **Protocol Number:** CA056-1107 **NCT Number:** NCT07215975 **Trial Status:** NOT_YET_RECRUITING **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Luspatercept (Reblozyl) **ATC Code:** B03XA06 **Disease Area:** \$\beta\$-thalassemia **Phase of Development:** Not Applicable (Observational / Real-World Study) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate luspatercept treatment in adults with transfusion-dependent beta-Thalassemia in the Middle East. **Secondary Objectives:** To assess real-world safety, reduction in red blood cell (RBC) transfusion burden, pre-transfusion hemoglobin levels, and disease-related best supportive care (BSC).

2.2 Background & Rationale To address the need for real-world evidence regarding luspatercept in the Middle Eastern adult population living with TDT. While clinical trials have proven that luspatercept promotes late-stage red blood cell maturation and reduces transfusion burden, this study evaluates its effectiveness and safety in routine clinical practice, capturing real-world dosing patterns and long-term outcomes.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Retrospective and Prospective) **Control Type:** Single-arm (Non-interventional) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Targeted Enrollment (\$N\$):** 200 evaluable patients **Number of Sites:** Multiple sites across the Middle East **Estimated Study Start:** 26-Feb-2026 **Estimated Primary Completion:** 28-Feb-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants of any race aged at least 18 years at time of initiation of luspatercept treatment. 2. Participants with documented diagnosis of transfusion-dependent β -thalassemia (TDT). 3. Participants who have been initiated on treatment with luspatercept as per the product's Summary of Product Characteristics (SmPC) no longer than 12 months prior to informed consent signature, and for whom therapy is ongoing. 4. Participants for whom the decision to prescribe luspatercept treatment is clearly separated from the physician's decision to include the participant in the current study. 5. Participants who have provided signed informed consent for participating in the study and for collecting and analyzing medical data pertinent to the objectives of this study.

4.2 Exclusion Criteria 1. Participants that meet any of the contraindications to the administration of luspatercept as outlined in the latest version of the locally approved SmPC. 2. Participants who are currently receiving or are planned to receive treatment with any investigational drug/device/intervention or who have received any investigational product within 1 month or 5 half-lives of the investigational agent (whichever is longer) prior to luspatercept therapy initiation. 3. Participants who are currently pregnant, breastfeeding, or planning a pregnancy during the study observation period. 4. Participants who have not provided signed informed consent for participating in the study and for collecting and analysing medical data pertinent to the objectives of this study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered subcutaneously as prescribed by the treating physician according to the locally approved SmPC for TDT. **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** Ongoing real-world treatment; prospective observation spanning up to the estimated completion date.

5.2 Concomitant & Prohibited Therapy Permitted: Routine RBC transfusions, iron chelation therapies, and standard disease-related best supportive care (BSC). **Prohibited:** Concurrent participation in another investigational therapeutic clinical trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Genotype Evaluation, Baseline Characteristics Baseline Day 0 Retrospective medical record review (transfusion burden,

prior hemoglobin levels) | | **Observation** | Prospective | Routine clinical visits: Transfusion logs, \$Hb\$ levels, Luspatercept dosing, AEs | | **End of Study** | Year 4 | Final chart review, Assessment of overall real-world effectiveness |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Change in transfusion burden. 2. Change in mean pre-transfusion hemoglobin level.

7.2 Secondary & Exploratory Endpoints 1. Change in transfusion-related visits. 2. Proportion of participants achieving $\geq 33\%$ reduction in red blood cell (RBC) transfusion burden (number of RBC units transfused) plus a reduction of ≥ 2 units. 3. Proportion of participants achieving $\geq 50\%$ reduction in red blood cell transfusion burden plus a reduction of ≥ 2 units. 4. Real-world dosing patterns and adjustments of luspatercept. 5. Disease characteristics describing age of diagnosis, genotype (β^0/β^0 , non- β^0/β^0), and splenectomy status. 6. Safety outcomes including incidence and severity of adverse events in a real-world setting.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted from routine clinical documentation and patient medical charts during the prospective period. **Serious Adverse Events (SAEs):** Reported according to standard post-marketing pharmacovigilance and real-world observational study guidelines. **Data Monitoring Committee (DMC):** N/A (Standard oversight for non-interventional studies).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled participants who received at least one dose of luspatercept and provided informed consent.

Sample Size Justification: Target sample of $N = 200$ is estimated to provide adequate descriptive statistics for the Middle East demographic. **Handling of Missing Data:** Handled via standard methodologies for real-world evidence (RWE), such as available case analysis without strict imputation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local regulatory guidelines. **Monitoring Plan:** Periodic source data verification (SDV) of site medical records against the study's Electronic Data

Capture (EDC) system. **Audit Trail:** Standard EDC locking procedures and data clarification workflows.

11. Study Identifiers & Governance

Primary ID: CA056-1107 **Registry Number:** NCT07215975 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** RELATE **Official Regulatory Title:** REal-World Application of Luspatercept in Adults With Transfusion-Dependent Beta-Thalassemia in the Middle East (RELATE): A Non-interventional Retrospective and Prospective Observational Study

12. Clinical Framework (Thalassemia Specific)

Primary Condition: Transfusion-dependent β -thalassemia (TDT) **Diagnosis Threshold:** Documented diagnosis of TDT requiring regular RBC transfusions **Medical Methodology:** Erythroid Maturation Agent (EMA) Therapy **Optimization Target:** Reduction in RBC transfusion burden and increase in endogenous hemoglobin

13. Study Procedures & Methodology

Management Pathway: Standard real-world clinical pathway for TDT **Education Protocol (HCP):** Administration according to locally approved SmPC guidelines **Education Protocol (Patient):** Standard clinical counseling regarding luspatercept therapy and transfusion needs **Quality Audit Mechanism:** Periodic tracking of real-world medical records and patient data registries

14. Observation & Follow-Up Schedule

Total Duration: Up to ~5 years (combining ≤ 12 months retrospective data and prospective observation up to 2030) **Onsite Visit Cadence:** Determined by the patient's routine standard of care schedule **Remote Monitoring Frequency:** Periodic clinical site data abstraction **Checklist Review Interval:** Routine EDC data entry intervals established by the sponsor

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				